

ActInf GuestStream 126.1 ~ Distributed Information and Computation in Ge

GuestStream | Distributed Information, Computation in Generic Quantum Systems | 58:38

hello greetings it is january 29th 2026 and we're in active inference guest stream 126.1 with chris fields james glazebrook and maurice defree so thank you all fellows for joining first chris is going to talk a little bit about the new book and then we'll have a discussion so if you're watching live you can write any questions in the live chat thank you chris to you uh thank you daniel um thanks for this opportunity for doing a brief presentation on this new book by jim glazebrook and myself on distributed information and computation in generic quantum systems this was published by springer last november i believe or december so is available online and what i wanted to start today by doing was just give a little motivation of what we're doing in this book um so that you can see once we get into it uh why we're we've assembled the topics that we've actually assembled so there are three main ideas here um the first is that quantum theory is not a very complicated theory about um the motions of small particles although that's how it's typically presented and it can certainly be used uh to describe um motions of electrons and atoms for example that's what it was developed to do from experimental motivations but in the last few decades it's been recognized that quantum theory is actually an information theory and so we're starting from the point of view that quantum theory is actually a theory of information exchange which means it's a theory of how systems prepare states and then measure those states and we'll see that running throughout the second main idea is that the free energy principle which many of you will be familiar with some maybe not which was developed 20 years ago as an idea in theoretical neuroscience uh our second idea is that the free energy principle is actually a practical criterion for deciding whether any bounded quantum system is persistent in time and again this is not a new idea to us in fact carl friston the originator of the free energy principle published this perspective and applied the free energy principle to essentially generic systems already in 2019 and the literature of the free energy principle since that time has been increasingly focused on the free energy principle as a foundation a foundational principle of persistence in physics and then the third basic idea is that active inference is a natural description of quantum computation active inference is the the picture of behavior that one obtains from the free energy principle active inference is a combination of learning and acting on the environment so as to change its properties its observable properties and if you think about an organism or a robot that can act on its environment then one of the primary kinds of actions it can take is just to move around to go somewhere else so if i don't like what my environment is doing to me here i can go to a different place and my environment in that different place may be different so by going on vacation for example i can alter how my environment is behaving to me so this is a very kind of um also fairly standard idea uh the term active inference was not introduced until

about 2010

and by i think carl fridsen again um but thinking of this as a way of describing quantum computation is somewhat new at least it um our motivation here is that any internal dynamics of a generic system can be described in this way as a combination of learning and action uh now clearly this rules out some things for example the idea of of a system that passively observes its environment is actually not formulaable within this theory it's an idea that doesn't make any sense uh because actions are always symmetrical across boundaries

as we'll see uh later on so a lot actually follows from these simple sorts of ideas uh some of which is in the book and of course it motivates our more recent work and some of this same thinking of course motivates the work of lots of other people besides us in the physics community and the biophysics community the cognitive science and ai communities etc so let's start with this idea that quantum theory is simple it's not about particles and waves it's about communication so what does that mean it means that what quantum theory describes are agents like us for example that set up experiments that make measurements and that communicate their results to other agents now this may be an odd way to think about quantum theory because very often quantum theory is associated with what's called the microscopic realm

but that characterization is far obsolete we know that quantum theory for example describes black holes which are some of the biggest things we know about in the universe

uh so quantum theory is actually a theory about exchanging finite quantities of information between systems and as we'll see uh it describes all systems as agents uh in a very clear sense the theory tells us how these agents work and what they can and cannot do with finite resources so for example

a large number of what are called no-go theorems follow from quantum theory many of them are in a sense variations

on girdle's incompleteness theorem they tell us that there are a number of things that quantum systems cannot do with information

the system that quantum theory now the the basis for this point of view is looking at the theory top down uh so instead of starting by postulating the existence of things like electrons for example uh we postulate this the existence of something that looks like a universe we ask what does it mean to say that a system must small thing like me is embedded in a big thing a universe and the only characteristic of this universe that's actually important is that it does not have an environment it contains everything

so in a very important sense it does not have a boundary because it does not interact with anything else it contains everything of interest

so the theory uh then allows us to address the question of how a generic system embedded in a generic universe interacts with the rest of the universe which is its environment

and it answers that question very clearly and cleanly and it also allows us to ask what a system can find out about its environment

so what sorts of information can a system extract from its environment

and how can it change its environment so what kinds of information can it transfer to its environment that causes some change in its environment state

as perceived by it
now when we think this way
uh it allows us to personalize the theory to recognize that it applies to us
and of course when one thinks of the theory as applying to oneself
one naturally connects it to concepts from places like cognitive science
concepts of computation of inference of evidence
uh and so on uh so uh part of this move to quantum information theory has been a move of connecting
physics to
cognitive science artificial intelligence neuroscience etc biology in general
and that's reflected in the marriage between quantum theory and the free energy principle
the theory of the free energy principle which everyone calls the fep
also asks a few simple questions but the first question it asks is why don't systems just dissolve into their
environments
why are there any systems that persist through time that are identifiable through time as the same thing
uh why doesn't the dynamics of the environment just rip the system to pieces
um so if systems don't dissolve into their environments then what do they have to do to prevent this
or just delay uh being dissolved into their environments and quantum theory has an answer to this question
that can be formulated very precisely
uh it says uh it says a system uh it says a system can delay being dissolved into its environment by
maintaining a weak interaction with its environment
by maintaining sparse coupling in the language of the fvp
yet by not allowing the environment to do anything really uh surprising uh or large
or damaging to it and the fvp formulates this in inferential terms the fvp says if a system wants to maintain
itself
it has to learn to predict what the environment will do and particularly what the environment will do to it
and it needs to learn how to act to avoid being dissolved by its environment and this is just active inference
that i
referred referred referred to earlier and it turns out that these two questions that these two answers to
the question about dissolution are basically the same that maintaining a weak interaction is what allows
learning and acting in a way avoids dissolution so it's weak interaction that allows the system to be an active
inference device so by putting these two concepts together it allows us to understand um what we regard as
interesting things and this is what we try to do in the book uh it allows us to understand how
any system divides its world into objects that can be described and manipulated so why is it that when we
look out at the world we see things like laptops and other people and tables and chairs and houses and
animals and plants and so on uh how does that work and in fact quantum theory lets us answer that question
for any system at all um how we can share objects and communicate by them about them so for example i'm
currently sharing this screen with you and we're talking about it how does that work and again
the formalism allows us to formulate that very clearly
it allows us to understand why we see objects and relations events as embedded in space-time

so why is it that when we look out at the world we see things organized in space and we see events happening in time those are very fundamental questions and uh quantum theory lets us answer them uh under the condition that we're talking about a persistent system so a system that complies with the free energy sensible and finally the theory lets us um say some very detailed things about what we or any system or any system cannot know about itself or its environment so these are the sort of questions that we take on in this book

and at this point i'm going to end this little slide presentation and

um give back to um

the main screen and actually bring up the book so that we can talk about it

so now let me do

um

thank you we see it great so um there's the book and we can just uh kind of

uh

cage through this and um talk about what's in it jim would you like to say a few things at this point

well let's see

where are we going to start yeah um

um

of quantum theory as a theory of particles to thinking of quantum theory as an information theory, which as I said, sort of started happening in about 1970 with people like John Wheeler and Richard Feynman as the major drivers, and is now a very large community, a very large effort, and it's what allows the concept of quantum computation.

So, it's a change of perspective that actually has enormous potential technological consequences and economic consequences.

I think we could go into chapter two, and the main points there...

include the concept of quantum reference frames, which permeates through a large part of the book.

And that starts in 2.4.

But Chris, do you want to comment on what comes before that?

Well, what comes before that is really just a discussion of how one can represent the relationship between any system and its environment.

And actually, let me just click up to this picture.

This is a picture that we've been using for several years now to understand how to represent the interaction between two generic systems, one of which we regard as the environment of the other.

And you can see that this is perfectly symmetrical.

So, we can switch these labels, system and environment, without changing anything.

And that, to look forward, has a large consequence.

It tells us that the free energy principle itself is completely symmetric.

And that, if we treat any entity as an agent, as the cognitive system, then we're forced to treat its environment as being an agent also.

So, a cognitive system.

So, that means that if I'm a cognitive system, to be consistent, I have to think of my environment as a cognitive system.

Because it's a system that I'm communicating with.

It's trying to talk to me.

And I'm trying to talk to it.

And we may be speaking very different languages, which we'll get to when we talk about quantum reference frames.

But just to focus on this picture in a minute, the theory allows us to describe the boundary between any two systems as just a collection of quantum bits.

And the systems interact by ultimately preparing and measuring these quantum bits.

So, this is literally a communication process with a finite number of information-bearing systems in between.

And this allows us to think of any interaction as communication over a channel in basically exactly the way that Shannon thought about it in the 40s.

Except here we're dealing with quantum processes, not classical processes.

And how it works.

Go ahead, Jim.

I was going to say, as a basic definition, for the sake of the audience, the quantum reference frame is a mapping from quantum states to a complex probability distribution.

And quantum reference frames incorporate a sort of a clock idea that involves memory.

And this is described in section 2.5.2.

And in particular, the diagram figure 2.4 on page 26.

I don't know if you want to leap forward to that now.

But that is one sort of embracing idea of the quantum reference frames.

Is this picture you're thinking of?

Or...

No, a 2.4 where you have basically the memory using a clock.

Oh, this picture.

Yes.

Yes.

I think that's an important picture from us.

And the fact is, as we go through descriptions of quantum reference frames and how they figure and paramount importance as far as the book is concerned.

To an extent, these descriptions are equivalent.

They can start off at this very basic generic level.

And then, as I said, a map between quantum states to complex probability distributions or a suitable framing of such to complex probability distribution.

But they can be tuned up from there to various levels.

And that we discuss later in the book, perhaps a bit too premature to discuss that now.

But one of the basic ideas driving these is the idea of reference and pointer states, which you'll find on page 28.2.6.

How you have this relationship between a system, a reference frame, and a pointer state.

Yeah, this is one of my favorite figures of all time, of course.

Let me jump in here for us.

Well, thanks, Chris, for producing it in the first place.

Maybe you'd like to comment on that.

I hope I'm not going too quickly through chapter two, but thinking that we have an hour at our disposal.

I think we should try to speed read in a sense.

So, Chris, maybe you could comment on that.

And that would bring us up to local operations, classical observers, which is also highly significant in what they're in now.

And what follows.

Yeah, let me give a brief kind of historical background here.

In classical physics, we have the concept of a reference frame, which is something like an X, Y, Z axis, for example.

And it was always viewed as an abstraction.

And in the 70s, businesses started to realize that reference frames had to be physically implemented.

And that means that all reference frames are quantum systems.

And so even things like rulers and clocks, which serve as reference frames, are physical systems that have physical properties that have to be taken into account when talking about measurement.

And it's only a short leap from there to realize that to use a meter stick of the scientist or the system using the meter stick, me, I have to have some notion of what length is.

So I myself have to be a reference frame for length or the little tick marks on the meter stick don't make any sense to me.

And similarly, if I'm going to read a clock, I have to actually be a clock.

Otherwise, the changes in the clock aren't going to mean anything about time to me.

If I have no sense of time, a clock is meaningless.

So it becomes clear that systems that operate as observers have to actually incorporate these reference frames into their dynamics.

And it's these internal reference frames that give our measurements meaning.

So with that comment, what physically is a reference frame, it's part of the dynamics of the system that you're working with.

What does the reference frame do?

It enables us to make measurements.

And what this picture illustrates is that the process of making a measurement always has two parts.

Physicists traditionally talk about measuring the pointer states of a system.

And that's a reference back to, you know, the gauges from the old days that actually had a pointer that moved.

And, but in talking about measuring the pointer states, one assumes that one knows what the system the pointer state is a pointer state of is.

But in fact, we have to ask, how did you find that out?

How do you know what system you're measuring?

And I like to use the example of sending a naive person into a crowded laboratory with lots and lots of stuff in it to read the value on a particular voltmeter.

Now, the first thing that person has to do is find that voltmeter.

So they have to go in the lab and look around and use some criteria that you've given them to find the thing that they're supposed to make a measurement of.

So to make a measurement, first one has to find the device, find the system, and then one can look at its pointer state.

And here I've just illustrated that with this old fashioned looking box that has a switch and two dials.

Before you can make sense of the pointers on the dials and the pointer on the switch, one has to actually find the device.

In an environment that typically is very cluttered, if we think about the environments that humans operate in or any other organisms operated in, they're full of things.

And we have to have a means of finding the ones that we're interested in if we're going to talk about the results of measuring those things.

So go ahead, Jim, if you want, we can move on to lock protocols and talk about.

Yes, I think so.

But I just want to say to kind of herald in what comes later, going back to figure 2.3, that we have a criteria for the possible

commutivity or non-commutivity of quantum reference frames, which is fundamental importance.

I thought I would mention that now because when Chris comments further, this commutivity or non-commutivity issue has a number of far reaching consequences.

And this even turns up in the case of LOCC, Local Operations Classical Communications, and various issues about the problem of sharing of quantum reference frames, QRF, observability, and LOCC, and the many questions of undecidability that we address here.

Maybe Chris, you can just give an outline of the basic idea following, say, 2.6, pages 30, 31.

Yeah.

Let me see if I can find that.

And we also, as you know, devoted.

Yeah.

Okay.

This is the basic idea of LOCC.

And we have an entire chapter in the book on this issue that Jim has brought up about whether reference frames commute.

And the phenomenon that it induces, which is typically called contextuality.

And it's probably best to understand that when thinking about two observers interacting with a joint environment.

That's what this picture shows in a very abstract way.

If you think of two observers, Alice and Bob, who are separate, they're interacting with an environment that they share, which here we label E.

And one can think of them communicating in two different ways.

One of which is to jointly manipulate some system.

So they may be manipulating a device of some kind.

In the very classical, or I shouldn't say classical, I should say canonical case for which LOCC protocols were originally described.

Alice and Bob are manipulating an entangled pair.

So two systems in an entangled state.

So they may be on opposite sides of the world manipulating this shared entangled state.

And at the same time, they're communicating classically.

They're saying, I'm making this kind of measurement.

I'm using this kind of device.

Make sure you're using this kind of device also.

And then they're sharing their results.

Alice may, in the case when they're sharing an entangled state,

and each making a sequence of measurements of copies of this state,

they will then afterwards share their results and discuss whether their results give them evidence of entanglement.

And that what counts as evidence of entanglement is formalized by statistical tests that were originally developed by Bell and Kogan and Specker back in the 60s.

So this is all very well established as technology.

And probably very soon we'll be interacting with things like our banks via protocols that involve sharing entangled states in this way.

Because they're communication protocols that are very resistant to eavesdropping.

So these kinds of protocols are extremely important technologically.

I think we could also add here, which is taken up later in the book, Chapter 7, I think, that any such protocol functions as a quantum error correcting code.

So this points us to a number of things which we do discuss later relative to quantum information, but also in questions of the nature of space-time and geometric phase.

Something we have taken up quite recently.

Anyway, as we say, that is discussed in greater depth in Chapter 7,

but I thought I would mention it now since Chris was talking about protocols, cryptography, and such things.

Well, actually, this diagram is a good place to introduce that.

I think so.

You can see here, yeah, we've depicted both of the communication channels as extending into the environment.

So think about what we're doing right now.

We are communicating via a physical system, which is the internet, as implemented by lots of fiber optics and lots of computers and lots of energy sources and so forth, that's extended around the planet.

And even though at this point, we're thinking what we're doing as classical communication, many of the devices in that internet depend for their operation on quantum theory working the way it does, such as all of the transistors and all of the computers, et cetera.

And so in doing this communication process, what we're doing right now, we are depending on this channel in our environment being reasonably free of noise, and it being able to correct errors that come up in the process, and all classical computer systems, all classical communication systems rely on various kinds of error correction, because all physical systems can fail.

And the environment at large is always acting on these physical systems that we use as communication channels.

For example, there are power failures and things like that.

And the internet is designed to cope with that.

So we rely in all of our communicative activities on the environment implementing ways of preventing errors from occurring in our communications.

And this is formalized in the notion of a quantum, first in the notion of a classical error correcting code, and then in the notion of a quantum error correcting code.

And a quantum error correcting code is just a way of preventing decoherence.

You may have heard the term decoherence, which means the destruction of quantum information, effective destruction of quantum information by spreading out too much in the environment, so that people can't use it.

So if our quantum channel has too much decoherence, there's what we will see as noise, because the information we're trying to share is leaking out into the environment and not being replaced. An error correcting code is a way of preventing leakage and replacing any information that does leak out. Yes, and connected with this is the relationship of QECC with entanglement entropy, which to an extent is behind what Chris is explaining.

That comes a bit later and that is significant in relationship to space-time, energy momentum, and one could think of a triangle relating entanglement entropy, energy momentum, Einstein's equation embraced by geometry.

But as I say, that comes later in the book.

Yes, you'll see we're still in chapter 2.

2.7, that discusses the no-go theorems for generic quantum systems.

In particular, we have theorem 2.7.1 there.

Do you want to comment on that, Chris?

No.

Let me see if I can find it here.

Yes, okay.

This is, as I said, we're still in chapter 2.

We're sort of, we're going, this chapter introduces the topics that the rest of the book is going to be about. So for example, error correcting codes, quantum communication pathways, emergent space-time, and so forth.

This last section in chapter 2 introduces the idea of no-go theorems.

The first no-go theorem was Bell's theorem, published by John Stuart Bell in the 1960s.

The second one was the Koch and Specker theorem, published by Koch and Specker, also in the 1960s.

And what Bell's theorem effectively says is that consistency with quantum theory requires us to give up one or both of two classical ideas.

One of the classical ideas is that everything happens locally.

And the other classical idea is that states that are not being measured, nonetheless have determinate values.

And those are both, as you'll recognize, kind of key notions of what we think of as the classical world.

Things happen locally.

And even if you're not looking at it, the moon is actually there.

The Koch and Specker theorem basically tells us that the context in which we make a measurement always matters.

And that by altering the context, one can alter the measurement outcome.

And these theorems are almost equivalent.

But since that time, since the 60s, a number of other theorems have been proven about what quantum systems can and cannot do and what they can and cannot know.

And this theorem that we put in Chapter 2 is a summary of some of the things that quantum systems, including us, can't determine by making observations.

So, for example, we can't tell by making an observation how our own reference frames work.

We're using them to make the observation.

So, by that process of using them, we cannot tell how they work.

We can't even tell what their input-output functions are.

So, if you think about that, one implication is that we cannot, by using language, fully understand language.

And that's very close to Gödel's theorem.

It's very close to points that Ludwig Wittgenstein made and other philosophers of language made in the middle 20th century.

So, the second part of this theorem is that an observer can't determine using a reference frame, i.e., by making an observation, the full properties of any other observational capability of that observer.

So, I can't, for example, using language, fully describe what my visual system does.

The third one here is that if I'm talking to you, I can't completely characterize the language you're using to talk to me.

And in a sense, that's obvious, right?

I can't see your internal processes.

I can just see what you tell me.

I can just see what's encoded on the boundary that separates us.

The boundary that we earlier represented by an array of cubits.

So, systems can't determine that they're sharing the same language.

And actually, that was a point that the philosopher Willard van Orman Quine made in the late 50s, in thinking specifically about language.

But this applies to any kind of measurement or communication protocol at all.

And this last one says that, again, that I can't completely determine your language, in a sense.

Two systems can't completely determine what each other are doing.

Now, Jim mentioned the entanglement entropy earlier.

Entanglement entropy is a measurement of how much entanglement there is in a system.

And the second theorem that we have here is, I think, one of the most important no-go theorems in the book.

It's a theorem that says that a system cannot measure the entanglement entropy across its own boundary.

So, for example, I cannot make a measurement that directly examines the internal state of my entire environment,

or the internal state of me, and compute the entanglement entropy between them.

No system can do that.

This has the consequence that no system can know that it has not, in fact, dissolved into its environment.

So, no system can know that it is, in fact, separate from its environment.

So, from a kind of cognitive science point of view, that's a very important theorem from physics, that systems can't determine that they're actually separate from anything else.

So, that's the end of this first section that sets up everything else in the book.

And let me just comment that the entire third chapter

lays out in great detail a category theoretic representation of how quantum reference frames work.

And it's this chapter that explains the conditions that Jim mentioned earlier for whether quantum reference frames commute.

And I think an interesting thing about this chapter is that it connects the physical notion of commutativity between operators,

which originally appeared in Heisenberg's uncertainty principle,

the lack of commutativity between position measurements and momentum measurements,

or energy and time measurements, to the mathematical notion of commutativity that's defined within category theory,

which is the idea of multiple paths leading to the same outcome.

And systems don't commute precisely when multiple paths don't lead to the same outcome.

So, you can see intuitively how communicativity relates to context dependence.

It's when you change contexts in midstream that paths will not lead to the same outcome.

So, Jim, we have about ten minutes to go.

Would you like to focus on any other particular section in here?

Well, I would say,

Chapter 4,

Chapter 4,

which is about the classical FEP,
Hierarchy or Bayesian influence,
and the Markov blanket concept is introduced there,
which we generalize in a sense to that of a holographic screen.
Maybe a lot of Chapter 4 is already familiar to the audience.

Chapter 5,
CCDs and Contextuality.

Well, that's basically about
the non-commutativity of CCDs
generates kind of quantum contextuality.
And we have our approach there.

I would say,
you know, six and seven
are really very significant,
and particularly when we advance these ideas to the notion of topological quantum field theory,
and from that incorporating spin networks into topological quantum neural networks.
So, there's much of the book left that might warrant a separate discussion.

Chris, Daniel, if you're interested.

Let me just throw a few.

Otherwise, it's a Herculean task for me in ten minutes to explain the rest of the book.

As what we're doing here.

Along with the appendices.

So, I think how far we've gone this morning is actually quite good,
because we've laid a foundation for studying quantum reference frames,
and how they're going to appear time and time again later,
how they are identified with topological quantum field theories,
and how they figure in LOCC,
and, as I said, the issues of problems of sharing the QRFs,
and to determine, well, what is the entanglement,
and generally these questions are undecidable,
if not impossible to tell,
and I suppose this is one of the basic mysteries of the subject,
and I think one tries to dig deeper.

It's a mystery of nature.

When Richard Feynman said,
nobody understands quantum theory.

He tried to work the theory out for himself using the path into the book's formulation.

I wonder if he was actually aware of these deeper issues like contextuality and problems of self-referentiality,

analogous to Godel's theorem,
that influenced Stephen Hawking's way of thinking in his latter years,
that physics is essentially incomplete,
and mathematics too because of Godel's theorem.

So, try to get to the point,
what is it that nature is telling us beyond these facets of the theory?

quantum theory,
non-locality, entanglement,
and such things that just end up being undecidable,
in many cases as far as we are concerned.

So, maybe that was the thing that Feynman had in mind,
what is it in nature that causes all of this?

And I don't as yet fully understand that.

I think it's a separate issue,
but let's see.

We've reached it now.

Danielle, how would you like to proceed?

Would you consider a follow-up discussion at some other time?

Yes, we can definitely do that.

Because there are a number of things there that will be of great interest to the active,
your active influence colleagues,
that very much relate to what we're saying here.

And we haven't really got onto the quantum free energy principle yet.

I think it's been discussed before.

It's there in the book if anybody wants to read it.

But I think that might warrant that chapter.

Some ensuing chapters would warrant a follow-up.

Chris, do you think it's a good idea?

Well, I think we might well want to do something that sort of picks up from what we've left off here.

And we can talk about the quantum version of the free energy principle
and how it distinguishes what's distinct between it and the classical version.

I mean, I think you'd say right here because it's the title of chapter five,
that the big distinction is that the classical free energy principle has no natural way of dealing with the sorts of
context shifts

that quantum theory introduces and that cause different measurement operators to be incompatible,
non-commutative.

And so a major motivation for introducing a quantum version of the free energy principle is to be able to deal
with these issues of contextuality and non-commutativity.

But to pick up on what you're saying, Jim, I think that doing something in the future that actually focuses on the relationship between communication, QECCs, and space-time, could be of great value here.

And I think you just gave a beautiful sort of explanation of Feynman's statement about quantum theory.

And I agree with you.

I think that what he was probably expressing was that one can study the world as much as one likes, and it's still surprising.

Yes.

And hard to understand.

You're going to keep getting surprised until you are not.

Oh, yes.

Constantly.

There's a saying that says we are learning until the day we die.

Thank you, fellows.

Really appreciate that.

Thank you, Daniel.

Thank you, Daniel.

This is very good of you to set this up.

Yep.

We'll talk about when to schedule it next and looking forward to people's questions.

So thanks again.

Bye.

All right.

Great.

Ciao.

Bye.

Bye.

it's over

very fun yeah

oh it's cool it's a wild

topic

yeah

yeah

yeah